

# Strain Measurement with Bonded Strain Gauges

Comparing quarter-, half-, and full-bridge Wheatstone configurations on an aluminum beam

|        |                                                                            |            |                                                                                                                                      |
|--------|----------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------|
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| Course | ME 343-004 Mechanical Lab I, NJIT                                          | Instructor | Prof. B. S. Mani · TA: Saba Mirahsani                                                                                                |
| Team   | Team 4: Luis Ramirez, Samuel Cohen, Edwin Ortez, Abhin Ghandi, Ramid Brown | Equipment  | Hollow 6061 aluminum beam, bonded strain gauges, gauge selector box, NI PXIe-4330 module, LabVIEW, weighted hangers, impact launcher |

## PROJECT SUMMARY

This experiment used resistance strain gauges bonded to a hollow 6061 aluminum beam to measure how the beam deforms under static bending, static torsion, and impact. The gauges were wired in quarter-, half-, and full-bridge Wheatstone configurations and read through an NI PXIe-4330 module in LabVIEW, with a 3 lb load applied at the free end. The bending load produced about  $-87\text{ }\mu\epsilon$  in the half-bridge configuration, and the torsional load produced about  $76\text{ }\mu\epsilon$  on a single quarter-bridge gauge. The comparison showed a clear link between bridge configuration and signal quality: bridges with more active gauges rejected noise and temperature effects far better than a single gauge.

## 1. Introduction

Strain measurement reveals how structures respond to load and is one of the most direct ways to validate theoretical models and confirm structural integrity. A strain gauge is a thin foil resistor whose resistance changes as it stretches or compresses. Because these changes are tiny, gauges are wired into a **Wheatstone bridge**, which converts small resistance changes into a measurable voltage. Using more active gauges in the bridge increases sensitivity and cancels unwanted effects such as temperature drift and electrical noise.

## 2. Theory

Table 1. Wheatstone bridge configurations (GF = gauge factor,  $R_L$  = lead resistance)

| Bridge configuration    | Active gauges | Strain from bridge output ( $V_r = \Delta V/V_{ex}$ )        | Typical use                             |
|-------------------------|---------------|--------------------------------------------------------------|-----------------------------------------|
| Quarter bridge (Type I) | 1             | $\epsilon = -4V_r / [GF(1 + 2V_r)] \cdot (1 + R_L/R_g)$      | Simple, single-point strain             |
| Half bridge (Type II)   | 2             | $\epsilon = -2V_r/GF \cdot (1 + R_L/R_g)$                    | Bending, with temperature compensation  |
| Full bridge             | 4             | $\epsilon = -2V_r / [GF(v + 1)]$ (Type I/II variants differ) | Highest sensitivity and noise rejection |

Theoretical strains follow from beam theory: bending stress  $\sigma = Mc/I$  with  $\epsilon = \sigma/E$ , torsional shear stress  $\tau = Tr/J$  with  $\gamma = \tau/G$ , and tip deflection  $\delta = PL^3/3EI$ . For a hollow circular section,  $I = \pi(d_o^4 - d_i^4)/64$  and  $J = \pi(d_o^4 - d_i^4)/32$ .

### 3. Procedure

Before each test, the gauge selector box was set to the bridge configuration for that loading case, and LabVIEW was configured with the input channel, excitation voltage (2.5 V), gauge factor (2.05), nominal gauge resistance (120  $\Omega$ ), sample rate, and offset nulling. For **bending**, a 3 lb weight was hung from the beam's free end and read with half- and full-bridge configurations. For **torsion**, the same 3 lb load was applied at a right angle to the beam axis using a moment arm and read with a single quarter-bridge gauge. For **impact**, a metal ball was launched through a tube onto the beam and recorded with a full bridge. Live strain was recorded in LabVIEW for each case.

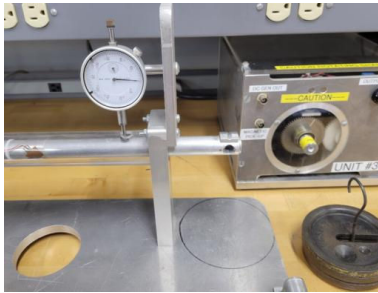


Figure 1. Beam, dial indicator, and impact launcher



Figure 2. Gauge selector box



Figure 3. Strain gauge interface

### 4. Results

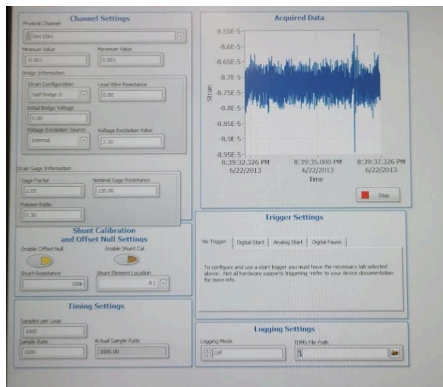


Figure 4. Half bridge (Type II), bending

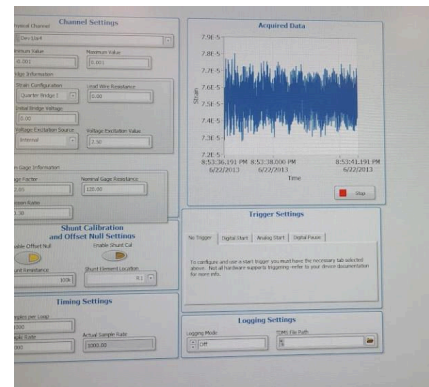


Figure 5. Quarter bridge (Type I), torsion

Table 2. Strain readings from the LabVIEW captures

| Configuration           | Load case     | Approx. mean strain                         | Signal band                 | Signal quality                             |
|-------------------------|---------------|---------------------------------------------|-----------------------------|--------------------------------------------|
| Half bridge (Type II)   | Bending, 3 lb | -87.5 $\mu\epsilon$                         | about $\pm 1 \mu\epsilon$   | Clean and stable, with an occasional spike |
| Quarter bridge (Type I) | Torsion, 3 lb | +76 $\mu\epsilon$                           | about $\pm 2.5 \mu\epsilon$ | Noticeably noisier                         |
| Full bridge             | Bending, 3 lb | $\approx -60 \mu\epsilon$ (from test notes) | —                           | Most consistent signal during the test     |

## 5. Discussion

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All three configurations produced a clear strain response consistent with the applied load, and the results show a direct relationship between bridge complexity and signal quality. With four active gauges, the **full bridge** gave the most consistent signal, because its arms cancel temperature effects and common-mode electrical noise. The **half bridge** produced a clean reading of about  $-87.5 \mu\epsilon$  under bending, with a noise band of roughly  $\pm 1 \mu\epsilon$ . The **quarter bridge**, with only one active gauge and no compensating arms, was the noisiest, with its torsion reading fluctuating between about 73 and 78  $\mu\epsilon$ ; any circuit-level disturbance passes straight to the output. Even so, it still clearly captured the torsional strain.

The negative sign of the bending strain indicates the gauges were on the compression side of the beam, while the positive torsion reading reflects the gauge orientation relative to the principal strain direction. Differences from ideal values likely come from small variations in the applied load and lever arm, calibration and offset nulling, and background electrical noise.

## 6. Conclusion

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The experiment showed a clear, repeatable relationship between mechanical load and measured strain in an aluminum beam under bending, torsion, and impact. Measurement stability improved as active gauges were added to the bridge: the full bridge gave the most consistent signal, the half bridge a clean bending reading, and the single-gauge quarter bridge the noisiest. For precise strain measurement, a bridge configuration with multiple active gauges is the better choice whenever the geometry allows it.

## References

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- Holman, J. P. (2012). *Experimental methods for engineers* (8th ed.). McGraw-Hill.
- Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.